Resourcing

Overview

Analyzing questions related to funding and constraints.

Research questions:
- How are OSPOs funded?
- What fraction of OSPO contributors are volunteers?
- What are the OSPOs' biggest blockers?

Relevant survey questions:
- How are your OSPO activities funded or allocated?
- Roughly how many people regularly spend time on OSPO activities without being formally resourced?
- Which of the following blockers has your OSPO encountered?

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tibble)
```

```
Warning: package 'tibble' was built under R version 4.4.3
```

```r
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```r
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```r
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```r
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
codebook <- read_excel(file.path(cfg$data_path, "codebook.xlsx"))
mc_simple <- read_tsv(file.path(cfg$data_path, "mc_simple.tsv"), show_col_types = FALSE, ski

ospo_funding <- read_tsv(file.path(cfg$data_path, "ospo_funding.tsv"), show_col_types = FALS
blockers <- read_tsv(file.path(cfg$data_path, "blockers.tsv"), show_col_types = FALSE, skip_
matrix_other <- read_tsv(file.path(cfg$data_path, "matrix_other.tsv"), show_col_types = FALS
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, s
```

How are OSPOs funded?

First, inspect answers from each org and check for disagreement from within the same org.

```r
ospo_funding$organization <- org_responses$organization

dupes <- ospo_funding |>
  group_by(organization) |>
  filter(n() > 1) |>
  arrange(organization, .by_group = TRUE)

# Keep commented for privacy!!
# dupes
```

As usual, there is some disagreement. Let's consolidate using my go-to heuristic: if anybody from that org said yes, then it's a yes from that org.

```r
ospo_funding_consolidated <- consolidate_org_responses_yn(ospo_funding)
```

```r
count_yes_responses(ospo_funding_consolidated)
```

```
# A tibble: 6 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    15 "External grant funding awarded directly to the OSPO"        external_g~
2    14 "Time formally allocated within people's existing roles"     formal_time
3    12 "Informal or \"side of desk\" time, not formally recognized" informal_t~
4     5 "Allocations from specific research projects or grants"      project_gr~
5     4 "Dedicated institutional budget for the OSPO"                dedicated_~
6     3 "Volunteer time outside work responsibilities"               volunteer_~
```

On average, how many resourcing sources does each OSPO have?

```r
# Convert y/n to 1/0
tmp <- ospo_funding_consolidated |>
  mutate(across(names(ospo_funding_consolidated)[-1], ~ if_else(. == "Yes", 1, 0))) |>
  select(-organization)

# Get mean number of yeses per row (organization)
mean(rowSums(tmp))
```

```
[1] 1.962963
```

Also inspecting the write-in answers.

```
# Keep commented for privacy!!
#sata_other$ospo_funding_other
```

What fraction of OSPO contributors are volunteers?

First, inspect answers from each org and check for disagreement from within the same org.

```
mc_simple$organization <- org_responses$organization
volunteers <- mc_simple |>
    select(volunteers, organization)

dupes <- volunteers |>
  group_by(organization) |>
  filter(n() > 1) |>
  arrange(organization, .by_group = TRUE) |>
  select(volunteers, organization)

# Keep commented for privacy!!
# dupes
```

Okay, great. I can see that only at one university did all respondents give a consistent answer. Similar to what I did in the "tools" notebook, I am going to just take the largest of the answers, because I expect that people are more likely to underestimate than overestimate on this one.

```
# Order of preference: keep whichever is further down the list
ordered_sizes <- c(
    "I don't know",
    "None",
    "1 person",
    "2 to 5 people",
    "6-10 people"
)

volunteers_consolidated <- volunteers |>
  mutate(volunteers = factor(volunteers, levels = ordered_sizes)) |>
  group_by(organization) |>
  slice_max(volunteers, n = 1, with_ties = FALSE) |>
  ungroup() |>
```

```
  mutate(volunteers = as.character(volunteers))


volunteers_consolidated |> count(volunteers)
```

```
# A tibble: 6 x 2
  volunteers        n
  <chr>         <int>
1 1 person          4
2 2 to 5 people    10
3 6-10 people       4
4 I don't know      3
5 None              5
6 <NA>              1
```

What are the OSPOs' biggest blockers?

This was a matrix question. I'll start by recoding the likert scale responses to numbers.

```
score_map <- c(
  "Not a blocker for us" = 0L,
  "Minor, we work around it" = 1L,
  "Moderate, it slows us down" = 2L,
  "Major, it limits our impact" = 3L
)

# Convert all text columns to numeric values
blockers_numeric <- blockers |>
  mutate(across(everything(), ~ score_map[.x]))
```

Hmm. I considered consolidating these to per-organization answers, but this feels like a pretty individualized question, so I'm actually not going to worry about multiple answers from the same org. We are just doing a survey of OSPO workers, and some of them happen to be from the same OSPO, but we're not going to consider that a concerning source of variation.

Also, I see that there are quite a few NAs. This can skew our averages. I'm going to report 2 numbers: severity (average of answers) and reach (how many people answered).

```r
summary_table <- blockers_numeric |>
  pivot_longer(
    cols = everything(),
    names_to = "challenge",
    values_to = "score"
  ) |>
  group_by(challenge) |>
  summarise(
    responses = sum(!is.na(score)),
    reach_pct = round(mean(!is.na(score)) * 100, 1),
    severity = round(mean(score, na.rm = TRUE), 2),
  ) |>
  arrange(desc(severity))

print(summary_table)
```

```
# A tibble: 10 x 4
   challenge         responses reach_pct severity
   <chr>                 <int>     <dbl>    <dbl>
 1 resources                30      90.9     2.53
 2 funding_timelines        32      97       2.06
 3 incentives               28      84.8     1.71
 4 awareness                32      97       1.62
 5 sponsorship              31      93.9     1.45
 6 tracking_impact          31      93.9     1.39
 7 policy                   28      84.8     1.32
 8 compliance               30      90.9     1.23
 9 tools                    30      90.9     1.23
10 culture                  30      90.9     0.83
```

Okay, so most of the questions had a response rate of >90%. That's good. In that case, I don't feel too bad about reporting severity only, as reach doesn't vary much.